MIL–O–39A
21 MAY 1957
SUPERSEDING
JAN–O–39
29 APRIL 1944

MILITARY SPECIFICATION

OARS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers oars.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT–S–176 — Sealer, Surface, Varnish Type, Floor, Wood or Cork.

MILITARY

MIL–V–1174 — Varnish, Spar, Water Resisting (Formula No. 80).

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Oars shall be made of wood. Wood shall be straight-grained, air-dried in accordance with commercial practice, and of any of the species of wood specified hereinafter (see 6.1).

(a) *White ash.* White ash includes green or Biltmore ash. Wood known commercially as "pumpkin ash", "swell-butted" or "brash" material will not be acceptable.

(b) *Hard maple.* Hard maple includes "sugar" and "black maple". Wood known commercially as "silver", "red", "soft" or "Oregon" maple will not be acceptable.

(c) *Red elm.* Red elm includes only slippery elm, ulmus fulva. White or American elm, or rock elm will not be acceptable.

3.2 **Shape and dimensions.** Shapes and dimensions of the oar shall be as specified in table I and as shown on figure 1.

FSC 2040

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TABLE I. *Dimensions*

A[1]	B	C	D	E	F	G	H	J	K	L	M	N	P	R
Ft.-In.	Inches	Inches	Inches	Inches	Inches	Inches	Ft.-In.	Inches	Ft.-In.	Inches	Inch	Inch	Inch	Inch
18-0	2¾	9	1½	2⁹⁄₁₆	2⁹⁄₁₆	6	4-8	5¼	6-0	1⅝	⅜	½	⅜	½
17-0	2¾	9	1½	2¼	2½	6	4-0	5¼	5-8	1⁵⁄₁₆	⅜	½	⅜	½
16-0	2¾	9	1½	2½	2	5¾	3-6	4¾	5-4	1¼	⅜	½	⅜	½
15-0	2¾	9	1½	2	1⅞	5¾	3-3	4⅝	5-0	1³⁄₁₆	⅜	½	⅜	½
14-0	2⅝	8½	1½	1⅞	1¾	5½	3-0	4½	4-8	1⅛	⅜	½	⅜	½
13-0	2⅝	8½	1½	1⅞	1¾	5½	2-9	4½	4-4	1¹⁄₁₆	⅜	½	⅜	½
12-0	2½	8½	1½	1¾	1⅝	5½	2-6	4½	4-0	1¹⁄₁₆	⅜	½	⅜	½
11-0	2½	6	1½	1⅝	1½	5½	2-3	4½	3-8	1¹⁄₁₆	⅜	½	⅜	½
10-0	2⅜	6	1⅜	1⁹⁄₁₆	1⁹⁄₁₆	5½	2-0	4½	3-4	1¹⁄₁₆	⅜	7⁄₁₆	⅜	½
9-0	2⅜	5	1⅜	1½	1⅜	5⅜	1-9	4½	3-0	1	⅜	⅜	⅜	½
8-0	2⅜	5	1⅜	1⁷⁄₁₆	1⁵⁄₁₆	5⅜	1-6	4½	2-8	1	⅜	⅜	⅜	½
7-0	2⅜	5	1⅜	1⅜	1¼	5⅜	1-3	4½	2-4	1	⅜	⅜	⅜	½
Tolerances plus or minus in inches														
1	⅛	½	⅛	⅛	⅛	¼	0	¼	½	⅛	⅛	⅛	⅛	⅛

[1] Nominal length, as specified (see 6.1).

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3.3 Weight of oars, specific gravity, weight per cubic foot.

3.3.1 Oars shall conform to both the weight of oars requirements and either the specific gravity or weight per cubic foot of wood requirements as follows:

3.3.2 *Weight of oars.* The minimum weights of oars at 12 percent moisture content shall conform to the schedule of weights specified in table II.

TABLE II. *Minimum weight*

Nominal length	Minimum weight[1]
Feet	*Pounds*
7	3⅝
8	4¾
9	6
10	7
11	9
12	10½
13	11¾
14	13¾
15	16
16	18½
17	20
18	22

[1] Minus ¾ pound for 7- to 12-foot oars having tapered handles; minus 1 pound for 13- to 16-foot oars having tapered handles.

3.3.3 *Specific gravity of oars.* The specific gravity, based on oven-dry weight and volume, shall not be less than 0.56 for all three kinds of wood as determined at both ends of representative oars.

3.3.4 *Weight per cubic foot of wood.* The weight per cubic foot of wood at 12 percent moisture content shall not be less than 36 pounds.

3.4 Straightness. Oars shall be straight. Oars shall be deemed to be straight within the meaning of this specification when:

(a) With the oar laid flatwise, the deviation from a straight line drawn from the middle of end of handle to the middle of end of blade shall not be greater than one-third of the diameter at the neck.

(b) With the oar held edgewise, a straight line drawn from the middle of end of handle to the middle of edge of blade shall not be outside of the loom of the oar.

3.5 Defects.

3.5.1 *Knots.* Two sound knots not over ⅜ inch in diameter will be permitted in the blade, provided they occur ¾ inch or more from the end. One sound knot not over $\frac{9}{16}$ inch in diameter will also be permitted in any part of the loom for a distance of 2 feet from the end, including the handle. The remainder of the oar shall be free from knots. Any knot on the handle shall be smooth to the grasp. Oars containing unsound, loose, encased, or pith knots and knot holes will not be acceptable.

3.5.2 *Checks.* Splits or openings in the surface of the wood if over $\frac{1}{32}$ inch deep will not be permitted.

3.5.3 *Worm holes.* Worm holes will not be permitted.

3.5.4 *Grain.* A slight cross grain will be permitted if the slope or pitch does not exceed 1 inch on 20 inches of length. Oars with knurls, curly grain, or compression failure in the looms will not be acceptable.

3.5.5 *Wane or bark.* Oars showing wane or bark will not be acceptable.

3.5.6 *Rot or dote.* Oars showing rot, dote, or incipient decay will not be acceptable.

3.5.7 *Shake and pith.* Oars showing shake or containing pith will not be acceptable.

3.5.8 *Color.* Oars shall be free from mineral streaks, except at handle extending not more than 15 inches from end, and for a similar length at end of blade, where it shall be

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limited to one fifth of the width. Calico ash mineral streaks, between these points, however variegated, will not be acceptable. Natural wood color will not be considered a defect. Discolorations caused by sap spiles (as in maple) or by other causes which do not impair the strength of the wood will be acceptable.

3.6 **Leather or copper bands.** Leather or copper bands shall not be furnished.

3.7 **Finish.** Oars shall be hand-dressed, smoothly and evenly finished, or machine-finished and sanded with sandpaper not coarser than 00. Oars shall be given one brush coat of water-resistance spar varnish conforming to Specification MIL–V–1174. As an alternate method, they may be treated with two brush coats or one 3-minute dip coat of sealer conforming to Specification TT–S–176.

3.8 **Workmanship.** Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Sampling for lot acceptance.**

4.1.1 *Inspection lot.* All oars of the same size offered for delivery at one time shall be considered a lot for purpose of examination.

4.1.2 *Sampling for visual and dimensional examination.* A random sample of oars shall be selected from each lot offered for Government inspection in accordance with Standard MIL–STD–105 at Inspection Level III for lots of 40 and under, Inspection Level II for lots of 41 to 800 and Inspection Level I for lots of 801 and over. The Acceptable Quality Level shall be 1.5 percent defective.

4.2 **Visual and dimensional examination.** Each of the sample oars selected in accordance with 4.1.2 shall be visually and dimensionally examined to verify conformance with all the requirements of this specification. Any oar in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective oars in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**

5.1.1 Unless otherwise specified in the contract or order, the oars shall be packed, reversing every other oar, in a solid end open wood crate, the minimum requirements shown on figure 2. In no case shall the angle between the edge members and the diagonals be less than 30 degrees nor more than 60 degrees. Suitable stuffing shall be applied between the tiers of oars at the ends to insure against warpage of the oars.

5.1.1.1 Each crate shall be made of well-seasoned lumber, free from knots or other defects which would tend to weaken the crate. Oars shall be packed in units as specified in table III.

TABLE III. *Packing of units*

Oar length	Number of oars per crate
Feet	
7	50
8	50
9	50
10	40
11	25
12	25
13	20
14	20
15	20
16	20
17	15
18	15

5.1.1.2 Each crate shall be reinforced with not less than 3 flat or round metal bands, one encircling the crate girthwise at each end over the end battens, and one encircling the crate girthwise equidistant from each end

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over the center battens. If flat bands are used, the size shall be not less than ¾ inch by 0.020 inch and if round wires are used, the gage shall be not less than 13. Filler cleats shall be nailed to the exterior side of battens between edge members to provide a flat, solid surface for stapling metal straps.

5.1.1.3 Each container shall be of the same uniform size, and shall contain the identical number of oars of one size. Containers shall be designed to fit the contents in a compact manner.

5.2 **Marking.** In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Nominal length (see 3.2, tables I and II).

(c) Species of wood required (see 3.1).

(d) Level of packing required (see 5.1).

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Navy—Bureau of Ships
Air Force

Other interest:
Navy—CGS

Preparing activity:
Navy—Bureau of Ships

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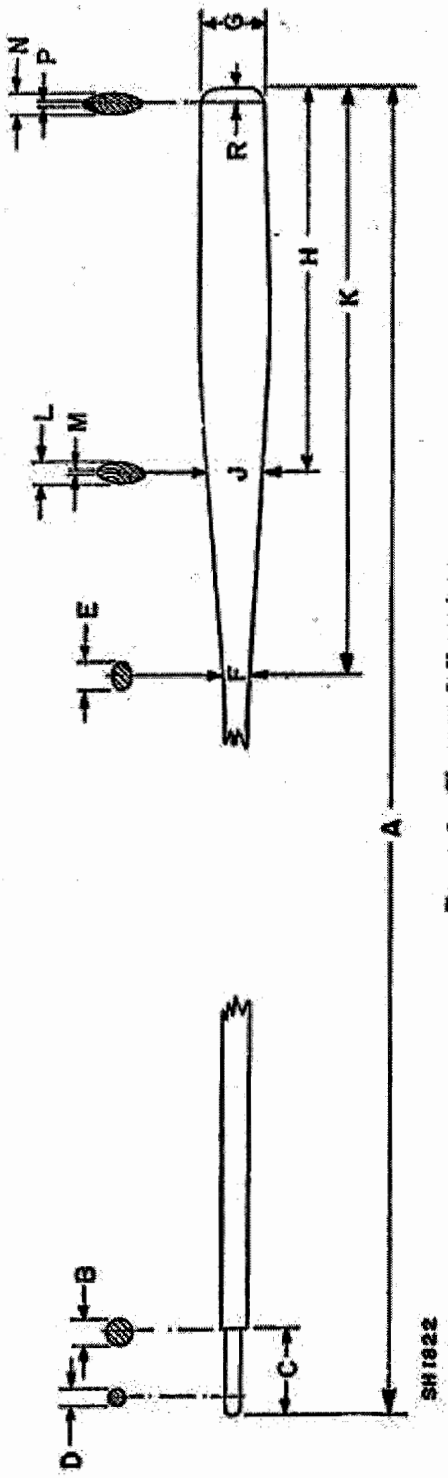


FIGURE 1. *Shape and dimensions.*

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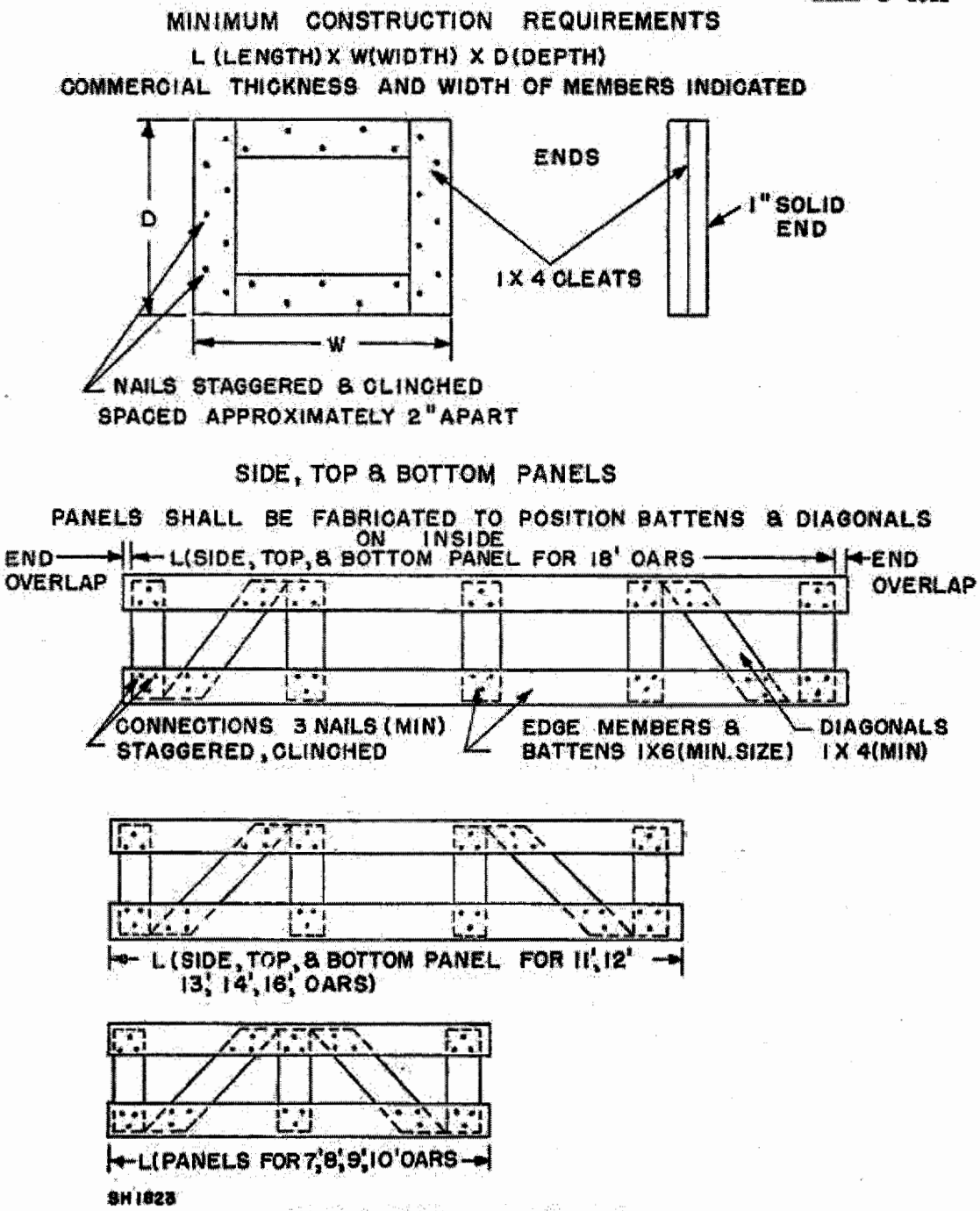


FIGURE 2. *Minimum construction requirements.*

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